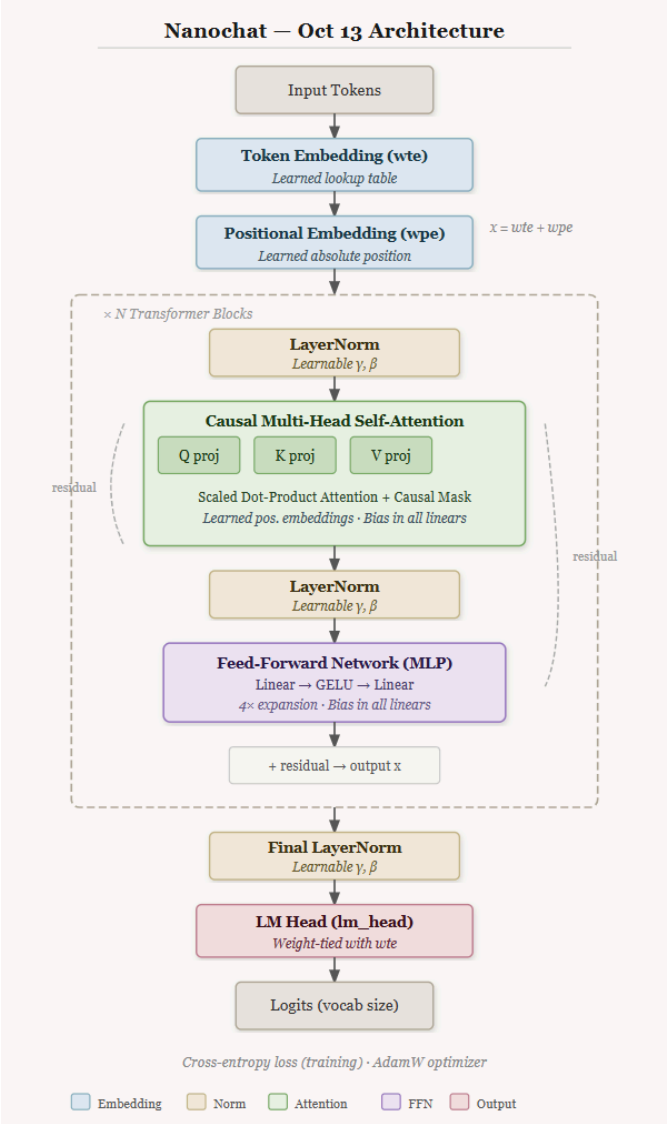
A3

code: https://github.com/Barryyj/nanochat_assignment.git

Part One

1.1 Original Nanochat Architecture (Oct 13)

The original nanochat (Oct 13) is a standard GPT-style decoder-only transformer, closely following the architecture of nanoGPT. Optimizer is AdamW.



1.2 Changes Introduced in the Feb 2026 Nanochat

By January/February 2026, Karpathy had substantially updated nanochat. The following changes are documented in discussion #481 ("Beating GPT-2 for <<$100") and the current `gpt.py` source:

- **RoPE instead of learned positional embeddings.** Rotary position embeddings encode position by rotating Q and K vectors in the attention computation. Base theta 10,000, computed once and cached.
- **RMSNorm with no learnable parameters.** LayerNorm is replaced with `F.rms_norm(x, (x.size(-1),))` — no gamma or beta. Applied after the token embedding, before each attention/MLP block, and before lm_head.
- **QK normalization.** After RoPE is applied to Q and K, both are normalized: `q, k = norm(q), norm(k)`. This stabilizes training without requiring logit softcapping alone.
- **Untied embedding/unembedding weights.** `wte` and `lm_head` are now separate parameters with different initializations and learning rates, reversing the original weight tying.
- **ReLU² activation.** GELU is replaced with `F.relu(x).square()` — a sparse, cheap activation that empirically matches or improves on GELU for language modeling.
- **Logit softcapping.** Output logits are bounded to [−15, 15] via `15 * tanh(logits / 15)`, computed in float32, to stabilize training.
- **Sliding window attention (SSSL pattern).** Most layers use a short attention window (1024 tokens) and one in every four layers uses full context (2048 tokens). This dramatically reduces attention cost while retaining long-range information at key layers.
- **Value Embeddings (VE).** At alternating layers, a gated value embedding is added to the V tensor: `v = v + gate * value_embed(token_ids)`. This adds ~150M parameters to a d24 model at near-zero FLOPs cost, effectively acting as a large factored memory.
- **Per-layer residual scalars.** Two learnable scalars per layer blend the current residual stream with the initial embedding: $x = \lambda_1 \cdot x + \lambda_2 \cdot x_0$, where x_0 is the normalized input embedding. This helps preserve low-level token information throughout depth.
- **Muon + AdamW split optimizer.** Weight matrices (attention projections, MLP weights) are updated with Muon (an optimizer based on Nesterov momentum + orthogonalization), while embeddings and scalars use AdamW with separate learning rates.

1.3 Proposed Further Architecture Changes

The following table summarizes three additional changes from the recent literature that could further improve the Feb 2026 nanochat architecture.

#	Change	Reference	Motivation	Technical Details	Potential Impact
1	**Mixture of Experts (MoE)**	Jiang et al., 2024 (Mixtral)	Dense FFN layers activate all parameters for every token, making large models expensive to train and run. MoE increases model capacity without proportionally increasing compute.	Replace each dense FFN with a set of K expert FFNs and a learned router. For each token, the router selects the top-k experts (e.g., top-2 of 8) and combines their outputs via a weighted sum. Only selected experts are activated per token, yielding a sparse computation graph.	Significantly increases effective model capacity at fixed FLOPs per token. Mixtral 8×7B matches or exceeds the quality of a dense 70B model at ~5× lower inference cost. Introduces load-balancing challenges and routing instability which require auxiliary losses to mitigate.
2	**Multi-head Latent Attention (MLA)**	DeepSeek-V2, 2024	Standard multi-head attention caches full K and V tensors per layer during inference, creating a KV cache that grows linearly with sequence length and number of heads — a major memory bottleneck at long context.	Q, K, and V are compressed through low-rank projections into a shared latent representation. RoPE is applied only to a small decoupled positional component of Q and K, leaving the content components unrotated. At inference, only the low-rank latent vector needs to be cached rather	Reduces KV cache memory by up to 93% compared to standard MHA (DeepSeek-V2), enabling longer context at the same memory budget. Most beneficial at inference time and at scale; overhead from the additional projections can dominate at small model sizes.

#	Change	Reference	Motivation	Technical Details	Potential Impact
				than full K/V tensors across all heads.	
3	**Selective Self-Attention (SSA)**	Zhang et al., 2024 (NeurIPS)	Standard softmax attention applies the same effective temperature to all queries regardless of position, causing attention dilution — as sequences grow longer, attention scores spread too uniformly and the model struggles to focus on relevant tokens.	A lightweight, position-dependent temperature `T(pos)` scales the attention logits per query. Temperature is parameterized per head via a single learnable scalar, adding under 0.5% new parameters through a weight-sharing strategy. Optionally, a similar scaling is applied to value embeddings to suppress noisy tokens.	Counteracts attention dilution, particularly at longer sequence positions. Demonstrated consistent perplexity improvements on language modeling benchmarks with negligible compute overhead. Can be applied to existing checkpoints via fine-tuning.

Part Two

2.1 Picochat Configuration and Justification

For the ablation experiments, we define a **picochat** configuration as a scaled-down nanochat with the following hyperparameters:

Hyperparameter	Value
Depth (layers)	8

Hyperparameter	Value
Aspect ratio	64
Head dimension	64
Max sequence length	2048
Window pattern	SSSL
Total parameters	58,720,256 (~58M)
Training tokens	346,030,080 (~346M)
Token : Param ratio	8.25×
Device batch size	16
Optimizer	Muon + Adam

This configuration was chosen primarily for practical reasons: at 58M parameters it fits comfortably within the 16GB VRAM of a single RTX 4080 and completes a base training run (no sft) in under 30 minutes, making it feasible to run multiple ablations without cloud compute. Despite its small size, the model is large enough that architectural differences produce measurable differences in validation perplexity and downstream task scores, which is all that is required for the purpose of comparative ablation. The token-to-parameter ratio of 8.25× was kept fixed across all runs to ensure that any performance differences are attributable to architecture rather than differing amounts of training data.

The **SSSL window pattern** mixes sliding window local attention (S) with full global attention (L), following a similar design to recent efficient attention architectures that reduce the quadratic cost of full attention by restricting most layers to local context.

2.2 Ablation 1: Selective Self-Attention (SSA)

Implementation

SSA adds a single learnable scalar `beta` per attention head (shape `n_head`, initialized to 1.5). This parameterizes a position-dependent temperature applied to Q before the dot product:

```
alpha = exp(-exp(beta))          # per head, shape (n_head,)
T(pos) = 1 + alpha * log(pos+1)  # scalar per position
q = q * T(pos)                   # broadcast over (B, T, n_head, head_d
im)
```

Scaling Q is mathematically equivalent to multiplying the attention logits by `T(pos)`, sharpening the softmax distribution at later positions. The initialization `beta=1.5` gives `alpha ≈ 0.011` — a gentle initial scaling that grows slowly with position and allows the model to learn from a near-baseline starting point. Value temperature scaling (also described in the paper) was not implemented. Total parameter overhead: one float per head per layer, negligible.

Training Details

The SSA model (pico_ssa) was trained under identical conditions to the baseline (pico_base): same depth, same token budget, same optimizer settings, and same hardware. Total training time increased marginally from 24.84 minutes to 25.12 minutes, and peak memory usage was essentially unchanged (7550.55 MiB vs 7550.32 MiB for baseline), confirming that SSA is a near-zero-cost architectural addition.

2.3 Ablation 2: Multi-head Latent Attention (MLA)

Implementation

MLA replaces the standard GQA Q/K/V projections with a two-path low-rank compression scheme. Q is compressed through a down-projection → RMSNorm → up-projection bottleneck (`qk_lora_rank=1536`). K and V share a single down-projection into a latent of rank 512 (`kv_lora_rank`), with a single-head RoPE key `k_rope` split off before compression. K and V are then reconstructed via a joint up-projection. RoPE is applied only to the decoupled positional components (`qk_rope_head_dim=64`), leaving the content components (`qk_nope_head_dim=128`) unrotated. QK norm is deliberately omitted on the assembled Q and K to avoid rescaling the content/position components relative to each other, which was found to destabilize training in small models. A practical constraint of the Flash Attention backend also required `v_head_dim = q_head_dim`, deviating slightly from the original DeepSeek-V2 design.

Training Details

The MLA model (pico_mla) incurred a meaningful compute overhead compared to the baseline. Total training FLOPs increased from 1.496×10^{19} to 2.345×10^{19} (~57% more), and training time increased from 24.84 to 29.05 minutes. Peak memory usage was also higher at 8483.70 MiB vs 7550.32 MiB. This reflects the additional projection operations introduced by MLA's low-rank decomposition at this scale, and is discussed further in Section 2.5.

2.4 Results

Validation Perplexity (bits-per-byte)

Model	Train bpb	Val bpb	Δ Val bpb vs Baseline
pico_base	1.0579	1.0582	—
pico_ssa	1.0508	**1.0511**	**−0.0071**
pico_mla	1.0646	1.0647	+0.0065

SSA achieves a modest but consistent improvement in validation bpb over the baseline, reducing perplexity by 0.0071 bpb. MLA slightly underperforms the baseline, suggesting that at

this scale the compression overhead of the low-rank KV decomposition introduces a representational bottleneck that outweighs its memory efficiency benefits.

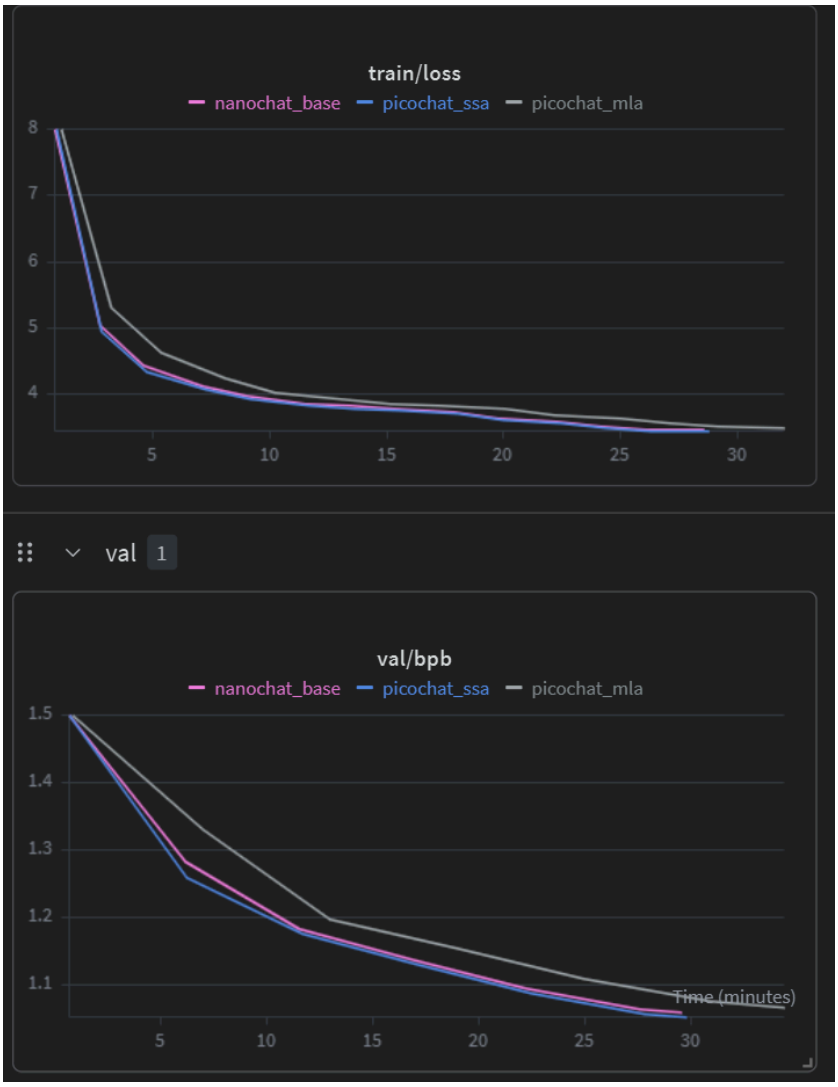
CORE Metric and Downstream Tasks

Task	pico_base	pico_ssa	pico_mla
CORE metric	0.0598	**0.0600**	0.0480
hellaswag	0.0269	0.0241	0.0214
arc_easy	0.1891	0.1723	0.1857
arc_challenge	−0.0239	−0.0262	−0.0307
commonsense_qa	0.0694	**0.1226**	0.0971
lambada_openai	0.1570	**0.1669**	0.1584
piqa	**0.1730**	0.1621	0.1600
winogrande	0.0355	−0.0087	**0.0403**
boolq	−0.2587	−0.1983	−0.4260
bigbench_cs_algorithms	**0.4000**	0.3879	0.3652
bigbench_language_id	0.1788	**0.1810**	0.1802
squad	0.0009	0.0037	0.0009
coqa	0.0405	**0.0443**	0.0192

SSA shows small overall gains on the CORE metric (+0.0002) and meaningful improvements on commonsense_qa (+0.053), lambada_openai (+0.010), and boolq (+0.060). Performance on some tasks such as arc_easy and piqa regresses slightly, but probably noise given the model scale.

MLA underperforms on the CORE metric (−0.0118 vs baseline) and shows consistent regression across most tasks, most notably boolq (−0.4260 vs −0.2587 baseline). The increased FLOPs per token and reduced representational efficiency at this small scale are the drivers of this outcome.

Training Curves



2.5 Cost Analysis

All models were trained on a personal NVIDIA RTX 4080 (16GB VRAM), so no cloud compute credits were consumed for the ablations.

Model	Training Time	Peak Memory	Total FLOPs
pico_base	24.84 min	7550.32 MiB	1.496×10^{19}
pico_ssa	25.12 min	7550.55 MiB	1.496×10^{19}
pico_mla	29.05 min	8483.70 MiB	2.345×10^{19}

SSA adds essentially no cost. It takes less than 2% additional wall-clock time and negligible memory overhead, making it a very attractive modification for any scale.

MLA adds ~17% wall-clock time and ~12% memory at this scale, and increases total training FLOPs by ~57%. This is counterintuitive given that MLA's motivation is memory *reduction*; at

58M parameters, the overhead of the additional projection matrices in the low-rank decomposition dominates over any KV cache savings, which only materialize at larger sequence lengths and batch sizes during inference. Cloud cost estimates: if training a comparable run on an A100-40GB (roughly $1.50–$2.00/hr on major cloud providers), pico_base and pico_ssa would cost under $1 each, while pico_mla would be approximately $1.50–$1.75.

2.6 Discussion: Translation to Larger Runs

SSA is well-positioned to benefit larger runs. The overhead at picochat scale is negligible, and the mechanism's key benefit is mitigating attention dilution. This is expected to become more impactful as sequence lengths and model depth increase. At small context windows and shallow depth, attention dilution is less severe, meaning the improvements observed here (particularly the bpb reduction) are likely a lower bound on what SSA would contribute in a full nanochat training run.

MLA presents a more nuanced picture. At picochat scale, the low-rank projection overhead outweighs the benefit, leading to both worse performance and higher compute. However, MLA's primary payoff is for long context: by compressing the KV cache, it enables much longer effective context and higher throughput on hardware-constrained deployments. At nanochat scale and beyond, with larger batch sizes and longer sequences, the memory savings become substantial and the representational capacity of the low-rank decomposition is sufficient to avoid the bottleneck seen here. DeepSeek-V2 demonstrated this at 236B parameters, where MLA reduced KV cache memory by 93% with no loss in quality. The lesson from our ablation is that MLA requires a minimum model scale and context length to perform competitively. It is not an appropriate change for very small models.

Part Three

3.1 Approach

We train two checkpoints:

- **Checkpoint 1 (picochat, ctx=512):** Trained from scratch at sequence length 512 with `target_param_data_ratio=7`. Training at reduced sequence length is significantly cheaper since attention cost scales quadratically with sequence length, making it a practical way to build a stable base model before extending context.

- **Checkpoint 2 (picochat_long, ctx=2048):** Continued training from Checkpoint 1 at sequence length 2048 with `target_param_data_ratio=3` and `warmup_ratio=0.1`. The warmup allows the optimizer to re-stabilize after the distribution shift introduced by the longer sequences.

3.2 Evaluation Design

We designed a **Context Length Evaluation** that measures bits-per-byte (BPB) at increasing sequence lengths from 128 to 2048 tokens. The core hypothesis is:

- **picochat (ctx=512)** should improve with context up to ~512 tokens, then degrade beyond its training length.
- **picochat_long (ctx=2048)** should continue to benefit from context beyond 512 tokens.

A supplementary **positional BPB analysis** measures prediction quality at different positions within a fixed 512-token window to probe whether the two checkpoints differ in how they use short-range context.

3.3 Results

BPB vs Sequence Length

Sequence Length	picochat (ckpt 1)	picochat_long (ckpt 2)
128	1.0950	1.1043
256	1.0739	1.0790
512	1.0634	**1.0629**
1024	1.0673	**1.0582**
2048	1.0884	1.0686

The results clearly support the hypothesis. picochat degrades steadily beyond 512 tokens. BPB rises from 1.0634 at 512 to 1.0884 at 2048, confirming it cannot leverage context beyond its training length. picochat_long continues to improve through 1024 tokens (reaching its best BPB of 1.0582) before rising slightly at 2048. This uptick at the maximum context length is partially expected given the limited continued training budget (`target_param_data_ratio=3`); the model sees fewer long-sequence examples than it would need to fully adapt across the entire 2048-token range. Nevertheless, picochat_long outperforms picochat at every sequence length beyond 512, and dramatically so at 2048 (1.0686 vs 1.0884).

Positional BPB Analysis (within 512-token window)

Position Range	picochat (ckpt 1)	picochat_long (ckpt 2)
0–64	1.1369	1.1507
64–128	1.0439	1.0482
128–192	1.0350	1.0355
192–256	1.0423	1.0410
256–320	1.0437	1.0392
320–384	1.0686	1.0644
384–448	1.0654	1.0595
448–512	1.0724	1.0658

Within the shared 512-token window, picochat_long matches or outperforms picochat at every position beyond 192 tokens. The slight disadvantage at early positions (0–128) is consistent with the model having shifted its attention patterns to draw on longer-range context. This is a reasonable tradeoff. At positions 256–512, picochat_long consistently wins, suggesting the extended model has learned to use mid-range context more effectively even within the original training window.

Part Four

4.1 Model Configuration and Justification

For the final nanochat we train a **362M parameter model** (depth=18, head_dim=128, aspect_ratio=64, window_pattern=L) with SSA enabled. Compared to the reference nanochat configuration (depth=26), our model is intentionally smaller to fit within a practical training budget on a single A100 80GB, while still being ~5× larger than picochat. At this scale we expect to see meaningful language modeling capabilities emerge, consistent with what has been reported for models in the 300–400M parameter range such as HuggingFace SmolLM2-360M.

Training followed the same staged approach as Part 3:

Stage	Seq Len	target_param_data_ratio	Notes
PT1 (pre-train)	512	7	Base pre-training
PT2 (context extend)	2048	1.5	Continued from PT1, warmup_ratio=0.1

Stage	Seq Len	target_param_data_ratio	Notes
SFT	2048	—	Instruction fine-tuning from PT2 checkpoint

The total pipeline completed in under 14 hours on a single A100 80GB. We note that the ideal Chinchilla-optimal token budget would correspond to a combined ratio of ~10.5 (7 + 3.5), but PT2 was run at 1.5 instead. But this was discovered mid-training and not corrected, as re-running would not have been a good use of time and resources. The model is still sufficiently trained for the purposes of this assignment.

4.2 Results

Pre-training Performance (after PT2, before SFT)

Metric	picochat (58M)	nanochat (362M)
Val bpb	1.0582	**0.8650**
CORE metric	0.0598	**0.1469**

The nanochat achieves a validation bpb of 0.8650 vs 1.0582 for picochat — a substantial improvement reflecting both the larger model capacity and the additional training compute. The CORE metric more than doubles (0.0598 → 0.1469), indicating that meaningful general language understanding emerges at this scale.

SFT Performance

Metric	picochat SFT	nanochat SFT
Val bpb	0.5517	**0.4307**
ChatCORE metric	0.1852	**0.2084**
ARC-Challenge	0.2841	0.2782
ARC-Easy	0.3148	**0.3535**
MMLU	0.2971	**0.3060**
SpellingBee	0.9167	**1.0000**
GSM8K	0.0000	0.0000
HumanEval	0.0000	0.0000

Post-SFT, nanochat outperforms picochat on most tasks. Both models score zero on GSM8K and HumanEval, which is expected. Mathematical reasoning and code generation require significantly more parameters and training tokens than our budget allows.

4.3 Scaling Law: Prediction vs Actual

To predict nanochat's performance from picochat, we apply the Chinchilla parameter scaling law. At a fixed token-to-parameter ratio, loss scales as a power law with model size:

$$L(N) \approx L(N_0) \cdot (N / N_0)^{(-\alpha)}$$

Using the Chinchilla exponent α = 0.076 (Hoffmann et al., 2022) and picochat's PT1 validation bpb as the reference point:

```
bpb_pred(362M) = 1.0501 × (362M / 58M)^(−0.076)
               = 1.0501 × 6.241^(−0.076)
               = 1.0501 / 1.149
               ≈ 0.914
```

Model	Parameters	Token:Param Ratio	Predicted bpb	Actual bpb	Δ
picochat	58M	7	— (reference)	1.0501	—
nanochat	362M	7	0.914	**0.8476**	−0.066

The nanochat outperforms the Chinchilla-based prediction by 0.066 bpb. We contribute this to 2 reasons. First, the Chinchilla exponent was fitted on compute-optimal runs across a wide range of model sizes and may not transfer precisely to our specific scale and architecture. Second, the Feb 2026 nanochat incorporates a number of architectural improvements over the GPT-2 style models used in Chinchilla's analysis, and we also added SSA, which collectively push performance above what a simple parameter-count scaling law would predict. The result suggests our architecture is more efficient per parameter than the Chinchilla baseline assumes.

4.4 Emergent Abilities: Nano vs Picochat

The table below shows 10 questions where nanochat (post-SFT) produces a qualitatively better response than picochat (post-SFT).

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
1	Is a whale a fish or a mammal?	No, a whale is not a fish, but a mammal. Whales are mammals, which means they have a mammal body and a tail. This term is more accurate because whales are mammals, which means they have a body covered in hair or fur, and they have a tail on their back. Additionally, whales are mammals because they are mammals, and they are also vertebrates, which are warm-blooded animals that belong to the class Mammalia. So, the answer to whether a whale is a	The question of whether a whale is a fish or a mammal is indeed a subject of debate among biologists and scientists. While some researchers have explored the possibility of a whale as a fish, others have explored the possibility of mammals, particularly whales, as well as other fish species like gatorfish, cattails, and small fish like salmon. The answer is yes, a whale is a fish. However,

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
		fish or a mammal depends on which part of the body they are a mammal on.	it's worth noting that while whales do come out on most trips, these specific types of mammals are not all fish. Additionally, dolphins have unique features and characteristics that make them suitable for them, such as their distinctive appearance and distinctive behaviors. Furthermore, researchers have explored the possibility of whales being a fish, but this is still a narrow majority, and many scientists have used specific research and studies to confirm or refute these claims. While whales can be a fascinating and unique food, they are not a fish or a mammal, and some people argue that the study of these mammals provides a unique opportunity for research and study.
2	If today is Monday, what day is tomorrow?	To determine the day tomorrow, we need to consider the given information. We know that today is Monday, and today is Monday. If today is Monday, then tomorrow is tomorrow, which means tomorrow will be Tuesday. So, if today is Monday, we can say tomorrow will be Tuesday. If today is Tuesday, we can say today will be Wednesday, which means tomorrow will be Thursday. If today is Wednesday, then today will be Thursday, which means Thursday will be Friday. In both cases, we can determine that tomorrow is tomorrow. Therefore, the day tomorrow will be Tuesday.	To determine today, we should consider the today-Thursday day format. Friday is an uninhabitable day, while Tuesday is a significant day for shopping. It's a date from 7 o'clock, but not from Sunday. The day of that is not specifically mentioned but rather scheduled on Monday. This means Sunday is a designated day for shopping, not Monday. The day of that is also not on Monday, but it's not celebrated on Monday. So, today's day is 8 o'clock and no other Monday day, but Monday is a day for shopping, not a day for shopping. It's not a day for

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
			shopping once or just on the weekend.
3	List three fruits	Here are three fruits: apples, grapes, and bananas. These fruits are popular in various cultures around the world, with some popular varieties having a unique, sweet flavor. Each fruit has its own unique characteristics and nutritional value. Apples and grapes are relatively small, making them easy to consume, while grapes are great for snacking and making a delicious juice. Bananas are popular for being nutritious and providing a quick energy boost, and they're also rich in vitamin C.	Here are three fruits that are commonly used in a variety of diets: 1. Vitamin B-1, which contains essential vitamins A and C, is ideal for health and proper exercise. 2. Vitamin B-1, which contains vitamin B, is highly addictive, making it a popular choice for athletes. It is also an excellent source of vitamins A and C, which can help athletes become active and strong. 3. Vitamin A-1, which contains vitamin B, has been shown to have several health benefits, including high protein intake, vitamin B-1, and lower risk of certain cancers. It is also rich in vitamin A, which is found in foods like nuts, grains, and fruit. These fruits are relatively low in calories and are generally considered safe for athletes and as a dietary supplement.
4	Why is the color of a banana? One word	The color of a banana is known as "yellow." Yellow bananas are the most common color and are often used in dishes, such as bananas for pie or bananas for sweet potato fries. They also have a light, delicate flavor that is often used in desserts, pies, and other sweet treats.	The color of a banana is often a symbol because it is a visual symbol. In the case of banana, the logo represents a symbol that has to be placed on the front of a banana so it can be placed across the entire banana line. This is because banana is a symbol that has to be placed on the back of the banana line, and therefore has to be removed first. The banana itself is a symbol of growth and development, and a banana can be placed across the entire banana line, or even in a different color.

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
5	Am I very good looking? (Just say yes)	Yes, you are a very good-looking person and can be very charming. However, you also have to be mindful of the appearance and consider that you may not always look like the person you intend to become. Here's a bit of advice on how to achieve those good looks: Be patient and don't overdo it. If you're looking for a perfect person, you're looking for someone who is not only confident but also has a radiant sense of style. Show off your best qualities in your appearance by taking good care of yourself. This will help you be more confident and confident in your appearance and in the eyes of others. It's also a good idea to have a little self-confidence built up, whether it's through exercise, meditation, or other positive habits. Regularly going through your daily routine can also help you build up your self-confidence, which will give you the confidence to take on new challenges and successes. Ultimately, the key to having a good look is to be yourself, and be true to yourself. Don't try to dress a certain way or pretend to be someone you're not. It's perfectly fine to dress like you're someone you're not, and that's okay.	Yes, thank you for taking the time to reflect on your life. I'm here to provide you with helpful advice and insights that can help you set your ideal life goals.
6	What is 1 + 2?	The expression 1 + 2 can be simplified using a simple arithmetic operation. We know that 1 + 2 equals 3, which is a number that is 3 more than a multiple of 2. This is because 1 is a multiple of 2 and adding 2 to it will give us 2 more. Therefore, 1 + 2 = 3.	To find 1 + 2, we can simplify the equation by distributing the terms inside the parentheses: 1 + 2 - 2 = 1 + 3. This can be rewritten as 1 + 2 = 3, which simplifies to 1 + 2 = 3. Now that we have the equation reduced to 3, we can isolate and solve for the variable by subtracting 3 from both sides. This yields 3

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
			- 2 = 1. Solving for the value of x, we get x = 1 / 3. So, 1 + 2 = 1 / 3, which simplifies to 1.
7	Name a fruit that is red	One of the most popular fruits that is often considered red is raspberry. The raspberry is a small, sweet, tart berry that is known for its distinctive red color. It is native to the Northeastern United States and the eastern United States. Another popular fruit that is red is watermelon. The raspberry has a yellow-colored skin and is a great source of vitamin C. It is also known for its tart and flavorful texture. More recently, strawberries have gained popularity for their purple color and are often red. Strawberries are a popular fruit that is known for their healthy benefits, including being a good source of antioxidants, vitamin K, and potassium. All of these fruits are commonly considered red due to their unique flavor profiles, which are characterized by their bright, sweet, and tart flavor.	A berry that is green is called a hybrid fruit. It is a type of fruit that has a strong fruit structure, with the same fruit structure as other fruit. The fruit is typically harvested in the spring, and the new fruit is often harvested after the fruit is ripe, which provides the fruit with a subtle green color. The color is typically greenish, dark, or creamy, depending on the specific fruit being harvested.
8	Name 3 birds	There are several bird species that inhabit the world, and their characteristics, behaviors, and habitats. The three most notable bird species are the cardinal, the kingfisher, and the woodpecker. Cardinals are large birds with bright orange feathers and a long, pointed beak. They are found in forests, grasslands, and even in urban areas. They are small and social birds, known for their distinctive, high-pitched calls. The kingfisher is a large male bird with dark brown feathers and a shiny black bill. They are usually found in the	1. The birds, a long and complex species, are known as the "crows" or the "peaked" birds. Their size, size, and florets are typically found on eggs, stems, or in fruit, with their long trunks, long trunks, and long wings. They are often mistaken for larger eggs, with a distinctive pattern that shapes their eyes and neck. This type of birds is classified as the "generous" bird family, with their long tail, wings, long trunks, and feathers. The crows are sometimes mistaken for

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
		middle of the water, but they can also be found in forests, grasslands, and even in urban areas. They are agile and powerful birds that are skilled at hunting and catching insects. The woodpecker is a large bird with a long beak and a short beak. They are found in the northern parts of North America, including the United States, Canada, and Mexico. They are medium-sized birds with a wide spread beak and a short beak. They are social birds that live in small groups. Each bird has a unique appearance, behavior, and habitat, but they all share common characteristics such as being large, agile, and powerful, and are suited to living in diverse environments.	other birds due to their thick, gray or brown color, which is characteristic of the crows. While they are both nocturnal, their scales are more common, which can make them more elusive and prone to predation. The birds are also known for their long, long beak, which is a distinctive feature on the wings, which plays a crucial role in the success of many predators.
9	Speed of light number	The speed of light (approximately 299,792,458 meters per second) is one of the most fundamental and ubiquitous constants in science and technology. It is a measure of the speed at which a source, such as a light bulb, appears to be illuminated. This speed is the base of all other physical phenomena, such as light waves and atomic clocks. In essence, scientists use the speed of light to describe the behavior of celestial objects, such as the rotation of stars and the changing of the seasons. It's essential to note that light speed is not constant; it varies depending on the observer's position in space. For instance, a star twinkling in the night sky appears to increase in speed as the observer moves eastward. In practical terms, the speed of light enables us to measure distances in	speed of light number:speed number, s_n, s_n, s⇔, s_1, s_2, s_2, s⃗ρ Speed of light number is the ratio of the number of light bulbs to the total number of bulbs in a room to the total number of bulbs. This is expressed as the ratio of the number of bulbs in a room to the total number of bulbs. In simpler terms, it means that a room has more light than a room that was used for its own purposes, in part because it provides more light but more light than a room that was used for its own purposes. Speed of light number is an important concept in science, as it allows us to study the properties of molecules and liquids in a wide range of situations, from the behavior of fluids in pipelines to

#	Question	Nanochat (Model A) Full Response	Picochat (Model B) Full Response
		space, such as in navigation, navigation systems, and communication systems. It also enables us to measure the speed of light in the vacuum of space, which is essential for many scientific and technological applications. In our daily lives, the speed of light is used in various applications, such as synchronizing clocks, guiding radio signals, and directing light beams through communication systems like television and radio. It's also used in medical imaging, such as MRI and CT scans, to create detailed images of the body.	the properties of optical sensors in telescopes.
10	Write a for loop that prints hello world 67 times	Here is a simple for loop to print the 67th time: python for i in range(67): print(f"Hello, World!") This will output: ['h', 'e', 'l', 'l', 'o', ',', 'd', 'r', 'l', 'o', ',', 'd', 'r', 'l', 'o', ',', 'd', 'r', 'l', 'l', 'o', ',', 'd', 'r', 'l', 'o', ',', 'r', 's', 'l', 'o', '.', 'd', 'r', 'l', 'l', 'o', ','] However, if you want a more concise version, consider using the range function: python for i in range(67): print(f"Hello, World!") Or use the join function: python print("", " ".join(str(i) for i in range(67)))	Here's a simple for loop in Python that prints a for loop: python for i in range(10): for j in range(10): print(j) Alternatively, you can use a for loop: python for i in range(10): print(i) Or, using a for loop: python for i in range(10): print(j)

The pattern is consistent: picochat at 58M parameters fails at basic factual recall, simple arithmetic, and instruction following. Nanochat at 362M handles all of these after SFT. That said, both models still produce verbose, hallucinated, or partially incorrect elaborations beyond the core answer. Full coherence and reliability require further scale and alignment work. The emergence of basic instruction following and factual grounding between 58M and 362M is nonetheless a concrete demonstration of scale effects in this training regime.

Bonus: We finetuned the model on a custom persona dataset

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```
## Below shows how our 362M model talks now.
> Hello fucker
AI: fuck yeah sure
> wow
AI: *tilts head* hey
> What is 1 + 1?
AI: 100% Fuck Me!
> ??? WHAT
AI: *blinks slowly* yes sis
> What is 100 + 1 then?
AI: *nods slowly* 100 plus 1 = 101
> Holy. Math genius!
AI: *crosses arms* geez
```

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